$PM_{2.5}$ AND PM_{10} CONCENTRATIONS FROM THE QALABOTJHA LOW-SMOKE FUELS MACRO-SCALE EXPERIMENT IN SOUTH AFRICA

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(Received 18 May 1999; accepted 20 December 1999)

Abstract. This article presents results from the particulate monitoring campaign conducted at Qalabotjha in South Africa during the winter of 1997. Combustion of D-grade domestic coal and low-smoke fuels were compared in a residential neighborhood to evaluate the extent of air quality improvement by switching household cooking and heating fuels. Comparisons are drawn between the gravimetric results from the two types of filter substrates (Teflon-membrane and quartz-fiber) as well as between the integrated and continuous samplers. It is demonstrated that the quartz-fiber filters reported 5 to 10% greater particulate mass than the Teflon-membrane filters, mainly due to the adsorption of organic gases onto the quartz-fiber filters. Due to heating of sampling stream to 50 °C in the TEOM continuous sampler and the high volatile content of the samples, approximately 15% of the particulate mass was lost during sampling. The USEPA 24-hr $PM_{2.5}$ and PM_{10} National Ambient Air Quality Standards (NAAQS) of 65 μg m^{-3} and 150 μg m^{-3}, respectively, were exceeded on several occasions during the 30-day field campaign. Average PM concentrations are highest when D-grade domestic coal was used, and lowest between day 11 and day 20 of the experiment when a majority of the low-smoke fuels were phased in. Source impacts from residential coal combustion are also found to be influenced by changes in meteorology, especially wind velocity. $PM_{2.5}$ and PM_{10} mass, elements, water-soluble cations (sodium, potassium, and ammonium), anions (chloride, nitrate, and sulfate), as well as organic and elemental carbon were measured on 15 selected days during the field campaign. $PM_{2.5}$ constituted more than 85% of PM_{10} at three Qalabotjha residential sites, and more than 70% of PM_{10} at the gradient site in the adjacent community of Villiers. Carbonaceous aerosol is by far the most abundant component, accounting for more than half of PM mass at the three Qalabotjha sites, and for more than a third of PM mass at the gradient site. Secondary aerosols such as sulfate, nitrate, and ammonium are also significant, constituting 8 to 12% of PM mass at the three Qalabotjha sites and 15 to 20% at the Villiers gradient site.

Keywords: chemical composition, domestic coal, low-smoke fuel, $PM_{2.5}$, PM_{10}

1. Introduction

Residential coal combustion has been identified as one of the major contributors to elevated ambient particulate levels in the industrialized areas of South Africa. To address public health issues, low-smoke fuels were developed as alternatives to coal in order to reduce smoke emissions.



Environmental Monitoring and Assessment **69:** 1–15, 2001.
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This $PM_{2.5}$ and PM_{10} (particles with aerodynamic diameters less than 2.5 and 10 μm, respectively) monitoring and source apportionment study forms part of the Qalabotjha Low-Smoke Fuels Macro-Scale Experiment in South Africa. Three low-smoke fuels (Flame Africa, Chartech, and AFC) were selected according to criteria established during a preliminary study in 1996 (Engelbrecht and Swanepoel, 1997). These fuels were combusted in domestic stoves and braziers by the residents of Qalabotjha over a 10-day period during the winter of 1997. Both the fuel testing and ambient monitoring programs were extended for 10 additional days to facilitate a complete phase-out of the D-grade domestic coal.

The Qalabotjha region is known to be subjected to strong temperature inversions (10 °C) in the winter months. During the field campaign, temperatures changed from –5 °C in the early mornings to 18 °C in the middays. The region has a typical 'Highveld' climate which is characterised by dry cold frosty winters and by warm summers with a moderate rainfall.

The township of Qalabotjha has approximately 15 000 residents residing in approximately 2500 dwellings. The dwellings range from informal galvanised iron shanties to formal brick houses. Although electricity is available, most residents prefer to use low-cost D-grade coal for cooking and heating. Ventilation is poor, so residents are constantly exposed to emissions from residential coal combustion.

The objectives of this article are to examine the $PM_{2.5}$ and PM_{10} mass and chemical compositions acquired in Qalabotjha and Villiers, and to compare results of gravimetric analysis on Teflon-membrane and quartz-fiber filters. Modeling of PM_{10} along with the source apportionment results are presented by Engelbrecht *et al*. (1999a, b).

2. Ambient Monitoring

Qalabotjha adjoins the north-eastern Free State town of Villiers and lies along the southern bank of the Vaal River (Figure 1a), close to the N3 route from Johannesburg to Durban, and approximately 150 km to the south-east of Johannesburg. This township was selected because of its size and its general upwind position in relation to the industrialized Vaal Triangle and Mpumulanga Highveld regions of South Africa.

Four sampling sites (shown in Figure 1b) were selected, including three urban sites in the black township of Qalabotjha (Qalabotjha Police Station, Crèche, and Clinic) where coal is used as the main energy source, and one gradient (Watson *et al*., 1997) site in the adjacent white community of Villiers where electricity is used. These four sites are within a radius of 1.2 km from each other to examine the spatial pollution variations within the neighborhood. The sampling sites were primarily selected to obtain representative measurements of study area; other considerations included the security of equipment, the availability of an uninterrupted power supply, and 24 hr accessibility by field operators.

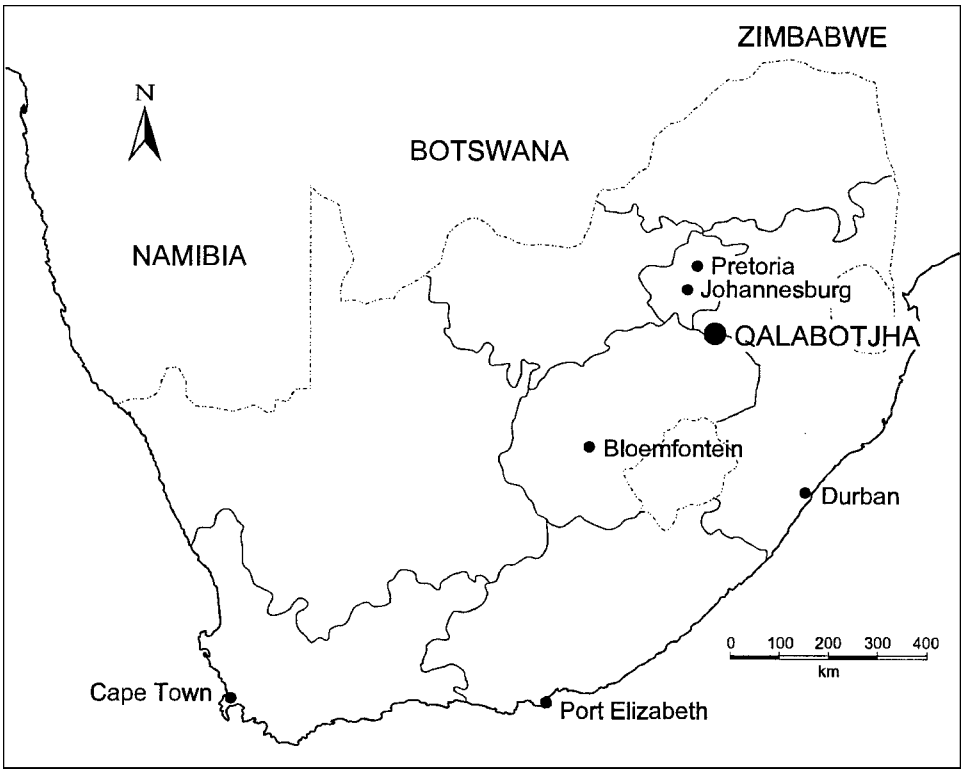


Figure 1a. Map of South Africa showing the Qalabotjha study area in relation to other geographical features.

A Rupprecht & Patashnick (Albany, NY) Tapered Element Oscillating Microbalance (TEOM) continuous PM_{10} sampler was placed alongside the integrated filter sampler on the roof of the Clinic. To conform with USEPA siting criteria, sampler inlets were placed at least 1.5 m above ground level (Chow and Watson, 1998) to eliminate the bias resulting from resuspended fugitive dust from unpaved roads.

Sampling systems shown in Figure 2 was configured with an Andersen Instruments (Smyrna, GA) SA246 PM_{10} size selective inlet, a URG (Waltham, MA) $PM_{2.5}$ cyclone, two manifolds, four filter holder sockets, four filter holders, four mass flow controllers, mass flow control box, vacuum pump, and a stainless steel enclosure. The flow through each of the mass flow controllers was set at 8.32 L min^{-1}, giving a total flow rate of 16.7 L min^{-1} through each of the size-selective inlets. Teflon-membrane and quartz-fiber filter substrates were mounted in specially designed filter holders (Engelbrecht *et al.*, 1993). Site visits and filter changes took place every 24 hr at approximately noon. The filters were handled in a dust-free room with a room temperature of 20±4 °C.

Daily, 24 hr per day sampling (noon to noon) was conducted from June 21 to July 20, 1997. Only the regular D-grade coal was combusted for the first 10 days of

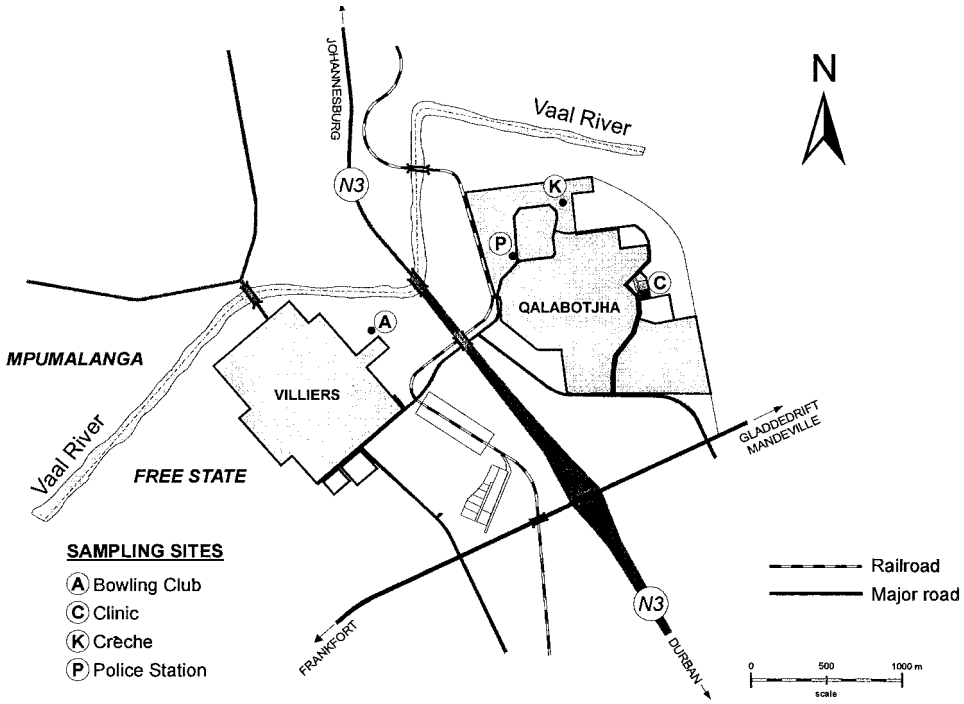


Figure 1b. Locality map showing the Qalabotjha and Villiers sampling sites.

the sampling campaign. Distribution of low-smoke fuels commenced on June 30, 1997, with the plan that the residents would be using these products exclusively from July 1 to July 10. However, low-smoke fuels were never completely substituted for the D-grade coal during this 10-day period since some residents preferred to exhaust the existing stocks of D-grade coal. The sampling program was extended for 10 additional days to facilitate a complete phase-out of D-grade coal.

3. Integrated Sampling

A total of 480 filters were exposed at the four sampling sites over the 30-day sampling period. Gravimetric analyses were conducted on the Teflon-membrane (Gelman Scientific, Ann Arbor, MI) and quartz-fiber (Pallflex Corp., Putnam, CT) filters. Due to the brittle nature of the quartz-fiber filters, special care had to be taken with the filter handling and weighing (Engelbrecht and Swanepoel, 1997).

Figure 3 compares the gravimetric concentrations for corresponding Teflon-membrane and quartz-fiber filter pairs. Regression statistics result in an x-intercept of –12.197, which implies that there is a constant loading on each of the quartz-fiber filters that is not found on the Teflon-membrane filters. This phenomenon had previously been encountered during the Vaal Triangle source apportionment study

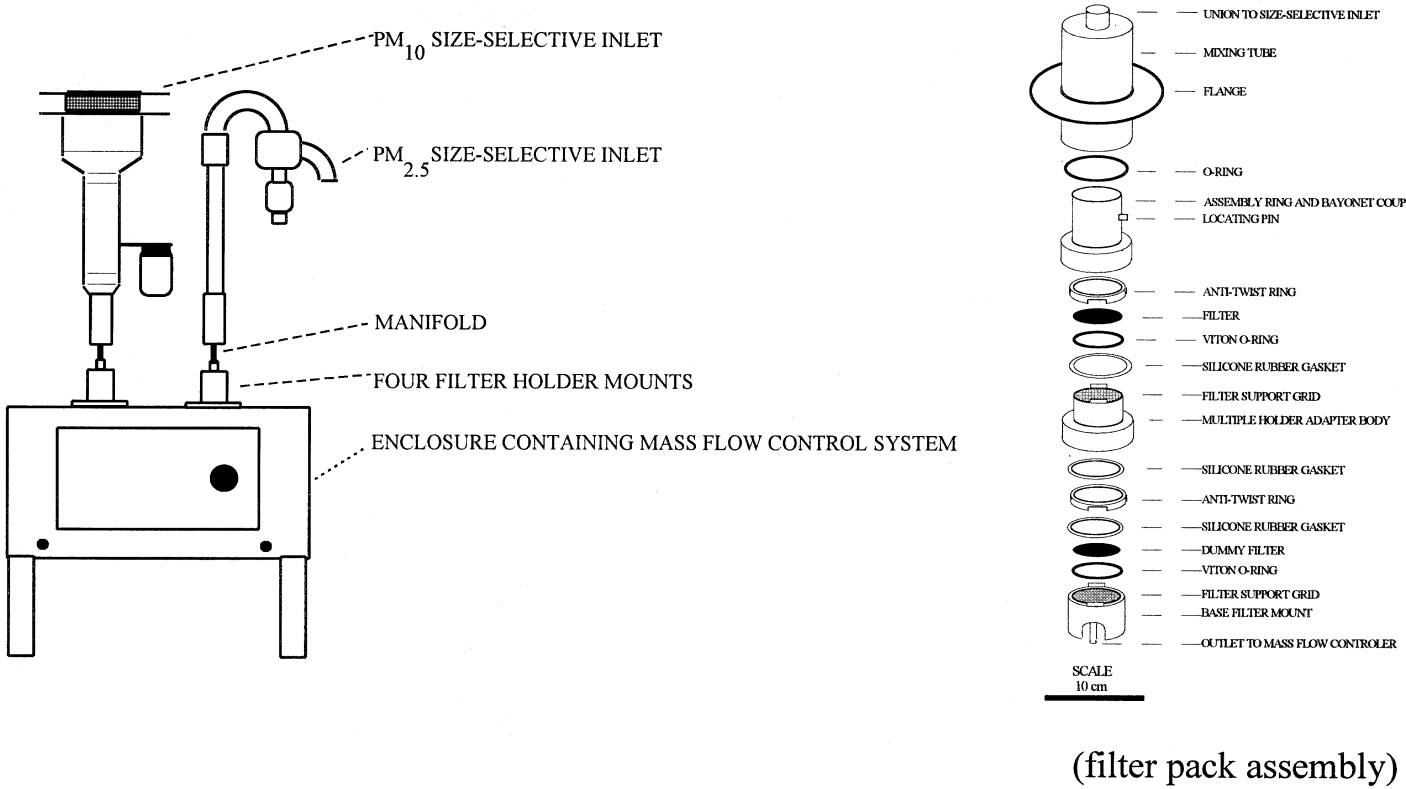


Figure 2. Integrated aerosol sampling system employed at Qalabotjha.

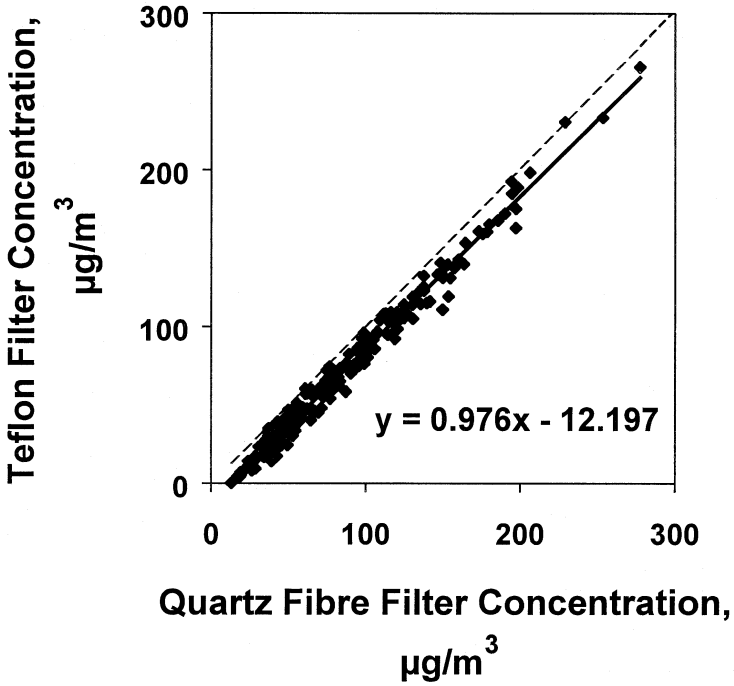


Figure 3. A comparison of the particulate masses on Teflon-membrane and quartz-fiber filter pairs. The dashed line indicates the one-to-one ratio.

(Engelbrecht *et al*., 1996; Engelbrecht *et al*., 1998). There are two possible causes for the 5 to 10% difference in gravimetric mass between the Teflon-membrane and quartz-fiber filters. First, Teflon-membrane substrates have a lower particulate loading capability, and particles may have fallen off of heavily loaded samples (‘caking’ effect) during filter handling or shipping. The quartz-fiber filters’ woven construction allows them to collect particles more homogeneously over a larger range of sample loadings. Second, mass ‘gain’ on the quartz-fiber filter may result from adsorption of organic gases prior to, during, and after sampling. Non-reactive Teflon-membrane filters minimize adsorption of organic gases. Corrections for adsorption can be made if field blanks were acquired during the field campaign. The following analyses are based on gravimetric results from the Teflon-membrane filters.

Figure 4 illustrates the temporal variations of PM concentrations at the four sites. Peaks and valleys in the data suggest the influence of meteorology during the study period. To a large extent, daily fluctuations can be ascribed to variations in wind speed and direction as well as number of hr during which it was windy. On windy days, Qalabotjha pollution levels approached those of at the gradient site (Bowling Club). Westerly winds affected the measured air quality at the Clinic site to the greatest extent. PM concentrations were consistently highest at the Clinic site, followed by the Crèche and Police Station sites, with the Bowling Club gradi-

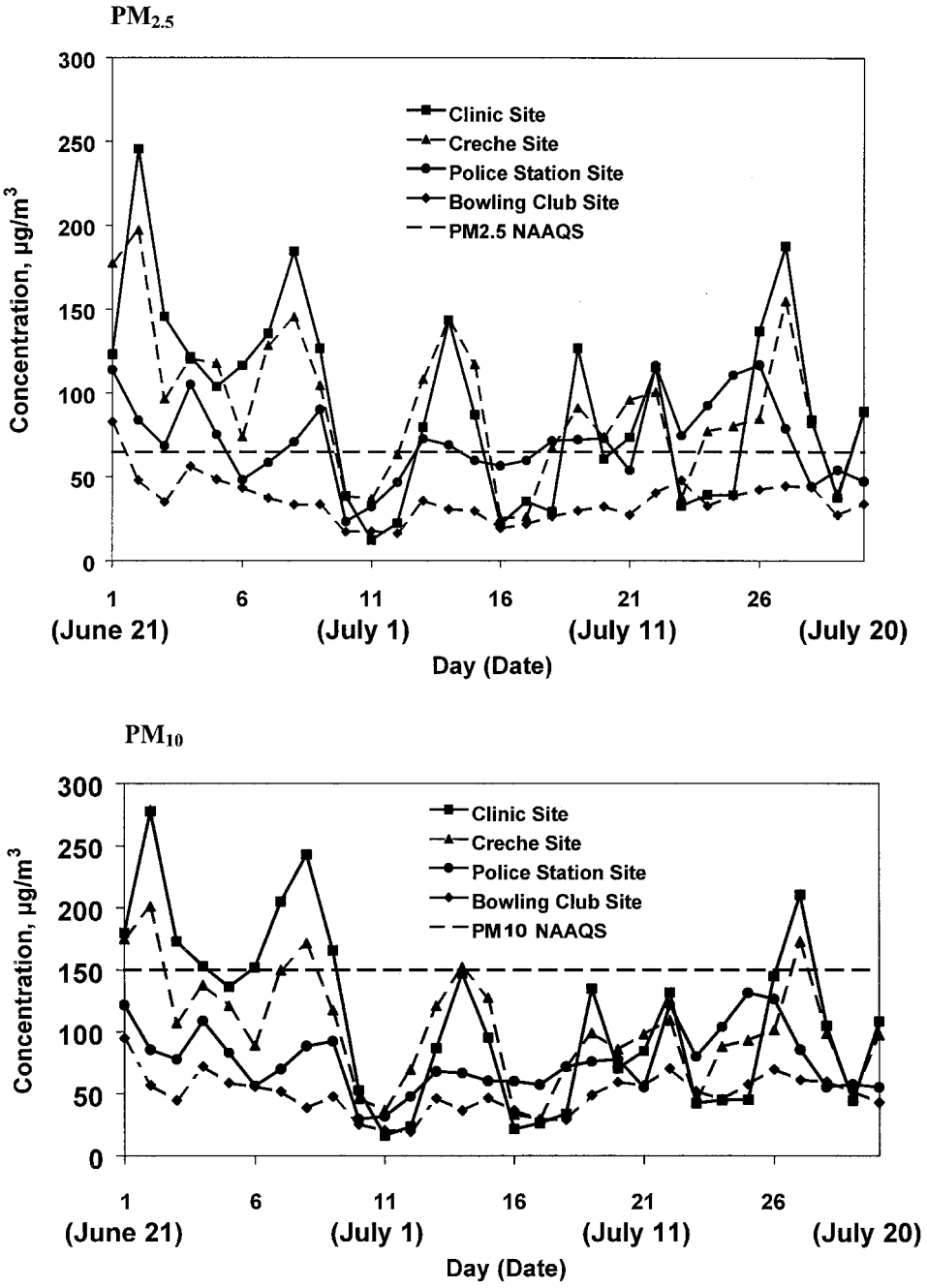


Figure 4. Gravimetric plots of the 24 hr $PM_{2.5}$ and PM_{10} particulate matter samples collected on Teflon-membrane filters between June 21 and July 20, 1997.

ent site being lowest. Pollution levels were highest during the first 10 days when D-grade coal was used, as compared to the remainder of the sampling campaign. The PM_{10} NAAQS of 150 μg m^{-3} was exceeded 24 times on 11 days, and the $PM_{2.5}$ NAAQS of 65 μg m^{-3} was exceeded 61 times on 23 days. The highest pollution level was reported on June 22, 1997 (day 2) with 278 μg m^{-3} of PM_{10} and 245 μg m^{-3} of $PM_{2.5}$.

Table I compares average $PM_{2.5}$ and PM_{10} concentrations among the three 10-day sampling intervals. Average PM concentrations were lowest between day 11 and day 20 of the experiment when a majority of the low-smoke fuels were phased in. At the Clinic and Crèche sites, average PM concentrations during the first 10 days when only D-grade coal was combusted were much higher than during the other two sampling periods. This was not the case at the Police Station and Bowling Club sites, where comparable results were found between days 1 to 10 and days 21 to 30. Thirty-day average $PM_{2.5}$ mass ranged from 71 to 93 μg m^{-3} at the three Qalabotjha sites, and was 36 μg m^{-3} at the Villiers gradient site. Corresponding average PM_{10} mass ranged from 77 to 112 μg m^{-3} at the three Qalabotjha sites, and was 49 μg m^{-3} at the Villiers gradient site. Large differences in PM mass were found between the Qalabotjha township and the adjacent Villiers community. Overall average $PM_{2.5}$ and PM_{10} concentrations were 86 and 97 μg m^{-3}, respectively, at the Qalabotjha sites, which are 50 to 60% higher than the corresponding measurements acquired at the Villiers gradient site, even though these sites are within 1.2 km of each other. Domestic coal combustion apparently had a significant impact on air quality in the residential neighborhood. Note that $PM_{2.5}$ constituted more than 85% of PM_{10} at the three Qalabotjha sites, and more than 70% of PM_{10} at the gradient site. Figure 5 shows that $PM_{2.5}$ to PM_{10} ratios decreased as PM concentrations exceeded 100 μg m^{-3}. Coarse particles (PM_{10} minus $PM_{2.5}$) may have been resuspended and become airborne as wind speed increased. Scanning electron microscopy analyses indicating the presence of silicon (crustal material in the coarse particles) and carbonaceous xenospheres (combustion particles in submicron particles), as shown in Figure 6, further confirms this hypothesis.

4. Continuous Sampling

A PM_{10} TEOM continuous sampler was collocated with integrated PM sampling systems at the Clinic site. Averages of 30–min PM_{10} concentrations were integrated over a 24 hr period for comparison. Figure 7 illustrates that TEOM PM_{10} mass concentrations followed a similar pattern to, but were consistently about 15% lower than, the corresponding values for the integrated sampler. Heating of the sampling stream to 50 °C in the TEOM resulted in evaporation of water vapor and semi-volatiles adsorbed on the particulates, which are enriched in emissions of coal combustion. Scatter plots (not shown) of PM_{10} mass between TEOM and

TABLE I

Summary of $PM_{2.5}$ and PM_{10} gravimetric results (micrograms per cubic meter) from Teflon-membrane filters

Type of residential combustion	Days									Mean		
	1–10			11–20			21–30			30 days		
	D-grade coal			Phase-in of low-smoke fuels						Mixture of D-grade coal and low-smoke fuels		
	$PM_{2.5}$	PM_{10}	$PM_{2.5}/PM_{10} \times 100\%$	$PM_{2.5}$	PM_{10}	$PM_{2.5}/PM_{10} \times 100\%$	$PM_{2.5}$	PM_{10}	$PM_{2.5}/PM_{10} \times 100\%$	$PM_{2.5}$	PM_{10}	$PM_{2.5}/PM_{10} \times 100\%$
Police Station	74	81	91%	61	62	98%	79	87	91%	71	77	92%
Crèche	120	131	92%	75	83	90%	84	95	88%	93	103	90%
Clinic	134	174	77%	62	66	94%	84	96	88%	93	112	83%
Mean, 3 Qalabotjha sites	109	129	84%	66	70	94%	82	93	88%	86	97	89%
Bowling Club	44	54	81%	26	37	70%	38	57	67%	36	49	73%
Mean, all four sites	93	110	85%	56	62	90%	71	84	85%	73	85	86%

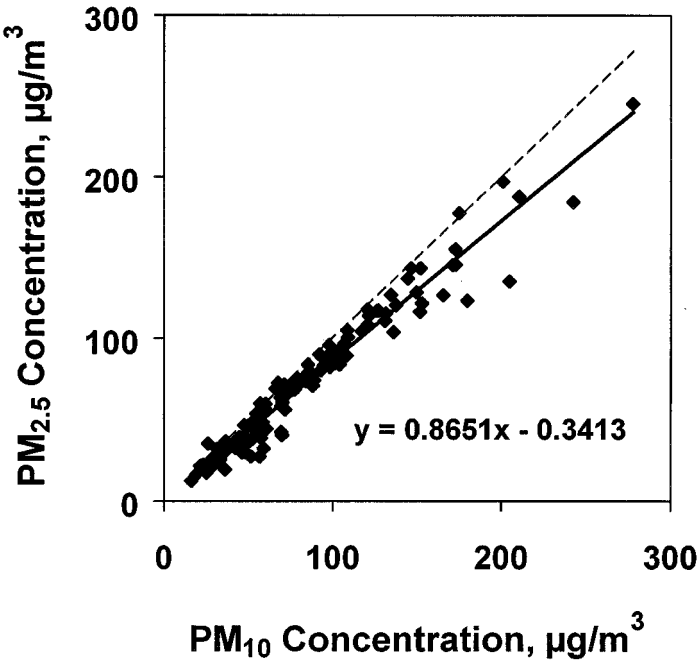


Figure 5. A comparison between the $PM_{2.5}$ and PM_{10} gravimetric results for Teflon-membrane filters. The dashed line indicates the one-to-one ratio.

integrated samplers indicate that as much as 45% of the initial mass on the TEOM could be volatilized.

Approximately 50% of the acquired samples with elevated mass were submitted for comprehensive chemical speciation. The selected 15 days included seven days from the first 10-day period, and four days each from the second and third 10-day periods. The samples were chemically analyzed following USEPA-approved guidance (Chow and Watson, 1998). The techniques employed included: 1) Energy Dispersive X-Ray Fluorescence analysis for 40 elements (Na, Mg, Al, Si, P, S, Cl, K, Ca, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, Ga, As, Se, Br, Rb, Sr, Y, Zr, Mo, Pd, Ag, Cd, In, Sn, Sb, Ba, La, Au, Hg, Tl, Pb and U) (Watson *et al.*, 1999); 2) Atomic Absorption Spectrophotometric analysis for water-soluble sodium (Na^+) and potassium (K^+); 3) Ion Chromatographic analysis for anions (Cl^-, NO_3^-, and SO_4^{2-}) (Chow and Watson, 1999); 4) Automated Colorimetric analysis for ammonium (NH_4^+); and 5) Thermal/Optical Reflectance analysis for seven fractions of organic and elemental carbon (OC1, OC2, OC3, OC4, EC1, EC2, EC3) (Chow *et al.*, 1993).

Table II summarizes average 24 hr $PM_{2.5}$ and PM_{10} and their major chemical constituents at each site. Similar results were found between the two size fractions except for crustal species (e.g., aluminum, silicon, calcium, iron). Since sample selection was intentionally biased toward days with higher mass, the averages in Table II are higher than the averages for all samples.

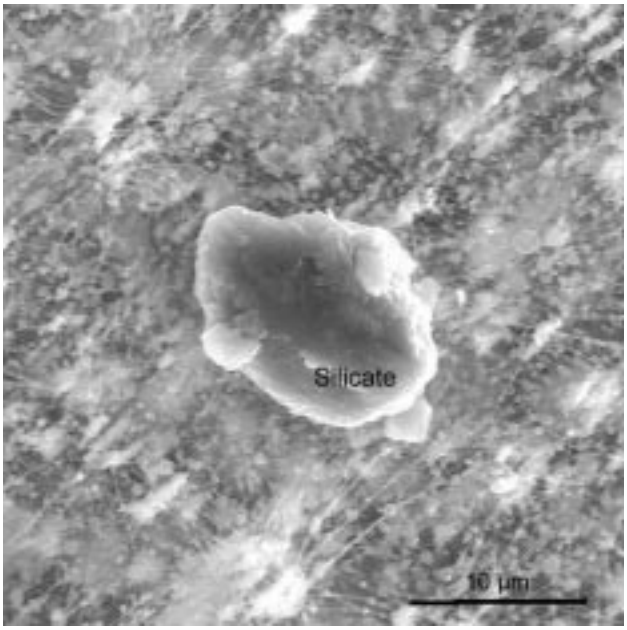


Figure 6a. Secondary electron image of silicate particle surrounded by smaller carbonaceous xenospheres (sample QALA345).

The most abundant PM components are organic and elemental carbon, accounting for more than half of PM mass at the three Qalabotjha residential sites, and for more than a third at the Villiers gradient site. Approximately 20 to 25% of total carbonaceous aerosol (non-carbonate carbon, sum of organic and elemental carbon) is organic carbon, reflecting the impact from coal combustion and vehicle emissions.

Secondary species such as sulfate, nitrate, and ammonium are also abundant, constituting 8 to 12% of PM mass at the three Qalabotjha sites and 15 to 20% at the Villiers gradient site. Spatial variations of these secondary species are relatively small, indicating its regional nature. Sulfate concentrations, in the range of 5 to 6 μg m^{-3}, are three times higher than the corresponding nitrate measurements.

The three most abundant PM_{10} crustal species are silicon (2.9 to 7.8 μg m^{-3}), aluminum (1.54 to 3.4 μg m^{-3}), and iron (0.7 to 1.9 μg m^{-3}). PM_{10} calcium concentrations are low, in the range of 0.35 to 0.78 μg m^{-3} at the four sites. $PM_{2.5}$ crustal species are less than one-tenth of the corresponding PM_{10} concentrations. This is consistent with the notion that a majority of crustal material is in the coarse particle size fraction. As wind speeds exceeded resuspension thresholds, geological-related particles became airborne and resulted in higher PM_{10} mass concentrations.

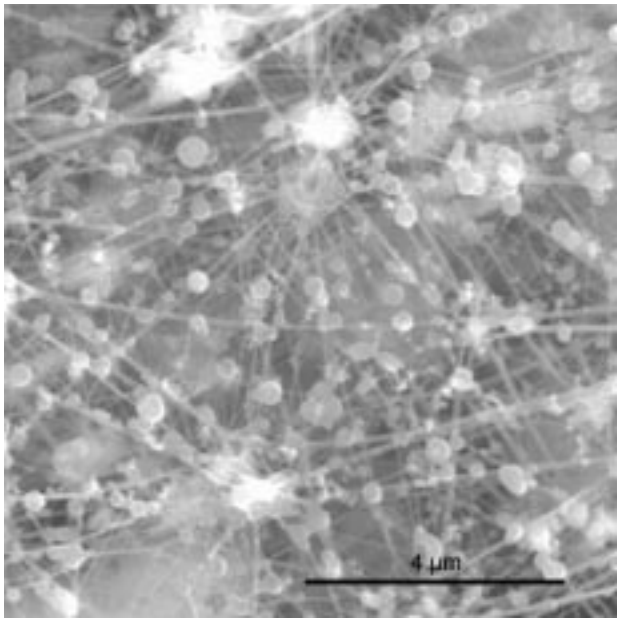


Figure 6b. Secondary electron image showing sub-micron sized carbonaceous xenospheres originating from the combustion of domestic coal and low-smoke fuels, on fibres and knots of the Teflon filter (QALA233).

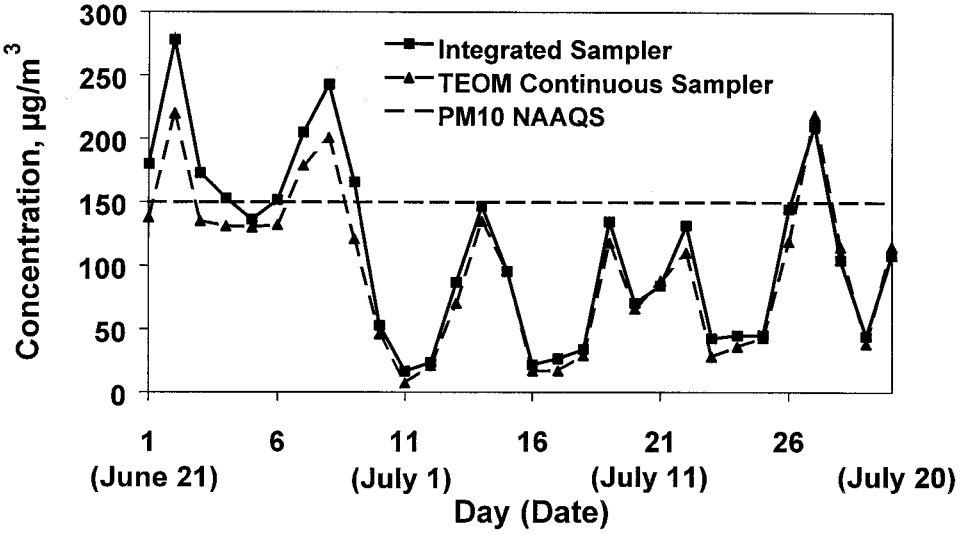


Figure 7. Day-to-day PM_{10} concentration variations from the integrated sampler and TEOM continuous sampler at the Clinic site, Qalabotjha.

TABLE II

Summary of $PM_{2.5}$ and PM_{10} mass and chemical compositions ($\mu g\ m^{-3}$)

Species	Size fraction	Police Station	Crèche	Clinic	Bowling Club
		(mean ± std. dev.)			
Mass	$PM_{2.5}$	84.257±20.491	127.299±32.042	134.814±45.380	41.857±12.314
	PM_{10}	89.783±23.465	138.159±33.573	154.276±51.560	5.299±12.784
OC	$PM_{2.5}$	32.467±12.302	58.687±17.392	62.937±23.374	11.651±5.310
	PM_{10}	32.741±12.003	60.724±17.595	65.362±25.852	12.578±5.091
EC	$PM_{2.5}$	10.553±3.041	16.292±3.593	15.327±4.805	4.001±1.586
	PM_{10}	10.172±3.285	16.481±3.697	15.317±4.572	4.263±1.288
NO_3^-	$PM_{2.5}$	1.751±0.819	2.100±0.921	2.280±1.049	1.107±0.874
	PM_{10}	1.878±0.849	2.200±0.949	2.628±1.275	1.402±0.838
SO_4^{2-}	$PM_{2.5}$	5.813±2.360	5.907±2.408	6.252±2.595	5.109±2.402
	PM_{10}	5.761±2.287	6.110±2.424	6.438±3.103	5.296±2.405
NH_4^+	$PM_{2.5}$	2.381±1.033	2.520±1.097	2.701±1.149	2.037±1.037
	PM_{10}	2.324±1.018	2.558±1.083	2.777±1.342	2.142±1.038
Mg	$PM_{2.5}$	0.051±0.037	0.027±0.021	0.032±0.028	0.060±0.027
	PM_{10}	0.087±0.039	0.067±0.053	0.202±0.119	0.162±0.082
Al	$PM_{2.5}$	0.171±0.043	0.209±0.058	0.265±0.094	0.218±0.085
	PM_{10}	1.410±0.471	1.903±0.693	3.424±2.045	2.028±0.842
Si	$PM_{2.5}$	0.419±0.117	0.490±0.158	0.686±0.233	0.583±0.238
	PM_{10}	2.936±1.000	3.935±1.321	7.847±4.715	4.813±2.315
S	$PM_{2.5}$	0.210±0.114	0.375±0.183	0.373±0.203	0.206±0.182
	PM_{10}	0.198±0.097	0.300±0.170	0.407±0.242	0.160±0.200
Cl	$PM_{2.5}$	0.276±0.225	0.718±0.319	0.797±0.470	0.018±0.029
	PM_{10}	0.414±0.278	1.062±0.442	1.371±0.777	0.060±0.068
K	$PM_{2.5}$	0.467±0.200	0.501±0.205	0.546±0.237	0.458±0.206
	PM_{10}	0.693±0.283	0.786±0.301	1.030±0.503	0.721±0.254
Ca	$PM_{2.5}$	0.078±0.029	0.086±0.033	0.102±0.038	0.112±0.050
	PM_{10}	0.348±0.121	0.420±0.158	0.783±0.479	0.617±0.285
Fe	$PM_{2.5}$	0.245±0.079	0.287±0.102	0.357±0.131	0.346±0.142
	PM_{10}	0.728±0.229	0.881±0.322	1.957±1.244	1.221±0.564

5. Summary and Conclusions

A 30-day field campaign was carried out during the winter of 1997 to evaluate the extent of air quality improvement when residents switched their source of cooking/heating fuel from D-grade domestic coal to low-smoke fuel. The USEPA

24 hr $PM_{2.5}$ and PM_{10} National Ambient Air Quality Standards (NAAQS) of 65 μg m^{-3} and 150 μg m^{-3}, respectively, were exceeded on several occasions during the 30-day field campaign.

Gravimetric measurements performed with Teflon-membrane and quartz-fiber filters are highly correlated, but resulted in a 5 to 10% higher mass on the quartz-fiber filters. This discrepancy may be due either to the caking effect (i.e., particle fall-off) of overloaded Teflon-membrane filters or to adsorption of organic gases on quartz-fiber filters.

Continuous PM_{10} mass acquired from TEOMs consistently reported 15% lower mass than the integrated aerosol samplers. Heating of the sampling stream to 50 °C in the TEOM resulted in evaporation of water vapor and semi-volatile compounds.

$PM_{2.5}$ constituted more than 85% of PM_{10} at the three Qalabotjha residential sites, and more than 70% of PM_{10} at the Villiers gradient site. Spatial variations of PM components are small except for crustal species. Carbonaceous aerosol is the most abundant component, accounting for more than half of PM mass at the Qalabotjha sites, and for more than a third at the Villiers site. More than 75% of the carbonaceous aerosol is organic carbon. Secondary aerosols such as sulfate, nitrate, and ammonium are also significant, accounting for 8 to 12% of PM mass at the Qalabotjha sites, and for 15 to 20% at the Villiers site. Sulfate concentrations, on average, are three times the corresponding nitrate concentrations.

PM mass concentrations were almost always highest during the first 10 sampling days when D-grade coal was used, and lowest between the 11th and 20th days when a majority of the low-smoke fuels were phased in. Average $PM_{2.5}$ and PM_{10} concentrations during the study period were 86 and 97 μg m^{-3}, respectively, at the three Qalabotjha residential sites, and 50 to 60% lower at the Villiers gradient site. Residential coal combustion had a significant impact on air quality in the vicinity of the residential neighborhood. Domestic fuel switching demonstrated potential to improve air quality.

Acknowledgements

The Mintek project team is indebted to the Department of Minerals and Energy for sponsoring the Qalabotjha macro-scale experiment; Mintek for its financial and other support; the Department Environmental Affairs and Tourism for the loan of the TEOM continuous sampler; the Villiers Town Council for their continued assistance with installing the sampling equipment; the community of Qalabotjha for their assistance and moral support; the Qalabotjha Police Station, the World Vision Crèche, the Qalabotjha Clinic and the Villiers Bowling Club for the use of their premises as sampling sites; and the Desert Research Institute (DRI) in Reno, Nevada, U.S.A. for conducting the carbon and other analyses of the filters at such short notice.

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